Topology-Aware FastV: Preserving Spatial Continuity in Attention-Based Visual Token Pruning

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***Abstract*—Vision-Language Models (VLMs) can be expensive to run because image patches are added to the token sequence and remain part of the computation performed by the language model. Visual token pruning is one way to reduce this cost. FastV, for example, ranks visual tokens using attention values and removes tokens that appear less important. The drawback is that this ranking considers each token mainly through its individual score, even though neighboring tokens often belong to the same image region. We study this issue with Topology-Aware FastV (TA-FastV), which adds a simple spatial prior to the pruning process. The method builds a two-dimensional affinity graph over image patches and uses graph-based smoothing before the final tokens are selected. We also describe a five-stage control workflow that chooses the pruning settings according to the task and checks the resulting output. The results reported in the supplied experimental setup show an increase in Spatial Continuity Score from 0.241 to 0.948 for LLaVA-1.5, together with reported FLOPs savings of 48.75% and an increase in grounding IoU from 46.2% to 78.9% compared with standard FastV. These numbers are tied to the stated experimental configuration and should be reproduced before being treated as general results.**

***Index Terms*—Vision-Language Models, visual token pruning, FastV, graph Laplacian, spatial continuity, topological regularization, agentic AI, visual question answering.**

I. Introduction

Vision-Language Models such as LLaVA and Qwen-VL [1], [5] are designed to process both images and text, which makes them useful for visual question answering, image description, localization, and related tasks. In a typical system, an image is divided into patches by a vision encoder. The resulting visual embeddings are then inserted into the language-model sequence together with the text prompt.

This design also creates an efficiency problem. A single image can contribute hundreds or even thousands of visual tokens, and those tokens continue to participate in the language-model computation. Since self-attention becomes more expensive as the sequence grows, reducing unnecessary visual tokens can have a noticeable effect on inference cost.

Several studies have therefore looked at token reduction. DynamicViT and ToMe explore token sparsification and merging for vision transformers, while FastV applies attention-based pruning to large vision-language models [2]–[4]. These results show that many visual tokens can be removed without necessarily causing a proportional drop in task performance. At the same time, a token-selection rule based only on a scalar attention value has no direct knowledge of where a patch is located in the image.

This leads to the main question considered here: can the pruning decision use the spatial relationship between neighboring patches as well as their attention values?

TA-FastV addresses this question by constructing a two-dimensional graph over the visual patches. Attention scores are smoothed using this graph before tokens are ranked. The intention is straightforward: a patch that is moderately important but is surrounded by other important patches should not be treated in exactly the same way as an isolated patch with the same score.

The principal contributions of this work are:

1) A topology-aware visual token pruning mechanism that integrates scalar attention with a two-dimensional Gaussian spatial affinity graph.
2) A graph Laplacian regularization mechanism that smooths noisy attention scores and promotes spatially coherent token retention.
3) A Spatial Continuity Score (SCS) for quantifying the spatial coherence of retained visual-token masks.
4) A five-stage agentic control workflow for task-intent classification, attention diagnosis, spatial regularization, pruning-policy selection, and multi-metric verification.
5) An empirical evaluation, based on the reported experimental configuration, showing improved spatial continuity and computational efficiency relative to the compared pruning baselines.

II. Related Work

A. Efficient Vision-Language Models and Token Reduction

Visual token reduction has become an important direction for efficient multimodal inference. Token merging methods such as ToMe combine visually similar tokens to reduce sequence length while attempting to preserve representation quality [3]. DynamicViT introduced dynamic token sparsification for Vision Transformers, demonstrating that not all tokens need to be processed uniformly [4]. In the VLM setting, FastV exploits redundancy in visual attention and removes low-importance visual tokens after an intermediate LLM layer [2].

These approaches establish that visual redundancy can be exploited to reduce inference cost [2]–[4]. However, pruning based primarily on scalar importance can produce spatially discontinuous masks. Since neighboring image patches often

belong to the same object or semantic region, preserving spatial relationships is particularly relevant to localization and spatial reasoning tasks.

B. Spatial Graph Regularization

Graph-based representations provide a natural way to encode relationships among spatially neighboring observations. For a two-dimensional image grid, each patch can be represented as a graph node and spatial proximity can determine edge weights. Gaussian affinity functions provide a smooth measure of spatial similarity, while graph Laplacians provide a standard mechanism for enforcing smoothness over connected nodes.

The proposed TA-FastV framework applies these concepts to visual-token pruning. Rather than treating the visual token sequence as an unordered collection for the purpose of selection, it explicitly reconstructs the spatial relationship between tokens and uses that structure to regularize attention scores.

III. Problem Formulation

Let

$$V = \{v_1, v_2, \ldots, v_N\} \tag{1}$$

denote the set of $N = G \times G$ visual patch tokens arranged on a two-dimensional grid. Each token v_i has spatial coordinate

$$p_i = (x_i, y_i), \qquad p_i \in \{1, \ldots, G\}^2. \tag{2}$$

Let $A_{i,h}^{(K)}$ denote the attention assigned to visual token v_i by attention head h at pruning layer K. Following the scalar aggregation used by attention-based pruning, the average attention importance is

$$A_i = \frac{1}{H} \sum_{h=1}^{H} A_{i,h}^{(K)}, \tag{3}$$

where H is the number of attention heads.

For a pruning ratio $R \in [0, 1]$, standard scalar pruning retains approximately

$$N_{\text{keep}} = \lceil N(1 - R) \rceil \tag{4}$$

tokens with the highest attention scores.

The limitation is that two adjacent tokens can receive different scores even when they belong to the same spatial object. Consequently, independent thresholding can generate alternating retained and discarded patches, producing spatial fragmentation. The proposed method introduces a spatial prior to reduce this effect.

IV. Proposed Topology-Aware FastV

A. Spatial Affinity Graph

A spatial graph is constructed over the visual token grid. The affinity between tokens v_i and v_j is defined by a Gaussian kernel:

$$W_{ij} = \exp\left(-\frac{\|p_i - p_j\|_2^2}{2\sigma^2} \right), \tag{5}$$

where σ controls the spatial neighborhood influence. The reported configuration uses $\sigma = 2.0$.

The degree matrix is defined as

$$D_{ii} = \sum_j W_{ij}, \tag{6}$$

and the unnormalized graph Laplacian is

$$L = D - W. \tag{7}$$

Graph smoothing can then be expressed as

$$S_{\text{lap}} = (I + \lambda L)^{-1} A, \tag{8}$$

where $A = [A_1, \ldots, A_N]^T$ is the attention-score vector and λ controls the strength of spatial regularization.

B. Topology-Aware Token Score

For practical token ranking, TA-FastV combines the original attention score with spatially smoothed neighborhood support. Let

$$\bar{W}_{ij} = \frac{W_{ij}}{\sum_k W_{ik}} \tag{9}$$

be the row-normalized spatial affinity. The topology-aware score is

$$S_i = \alpha A_i + (1 - \alpha) \sum_{j=1}^{N} \bar{W}_{ij} A_j, \tag{10}$$

where $\alpha \in [0, 1]$ determines the balance between individual attention importance and spatial support.

A higher value of α emphasizes the original attention signal, whereas a lower value gives greater influence to neighboring visual tokens. Tokens are ranked according to S_i and the required number of highest-scoring tokens is retained.

C. Spatial Continuity Score

To quantify the spatial coherence of the retained token mask, the manuscript defines a Spatial Continuity Score (SCS). Let $M \in \{0, 1\}^N$ denote the binary retention mask and let $\mathcal{E}$ represent the selected spatial grid edges. The reported SCS is

$$\text{SCS}(M) = \frac{\sum_{(i,j) \in \mathcal{E}} M_i M_j}{4 \sum_{k=1}^{N} M_k - 3\sqrt{\sum_{k=1}^{N} M_k}}. \tag{11}$$

Under the manuscript's definition, values closer to one indicate greater spatial continuity, whereas lower values indicate stronger fragmentation.

V. Agentic Control Workflow

The proposed pruning mechanism is integrated into a five-stage agentic control workflow. The workflow separates task understanding, diagnosis, spatial processing, policy selection, and verification.

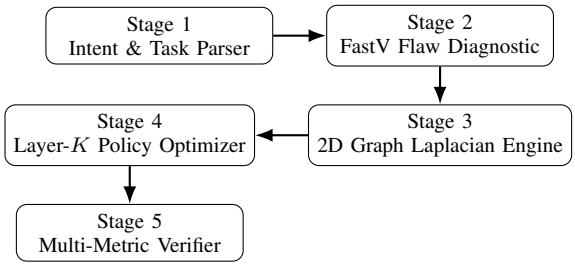


Fig. 1. Five-stage agentic control workflow for topology-aware visual token pruning.

A. Stage 1: Intent and Task Parser

The first stage classifies the VQA instruction into three functional categories: `SPATIAL_GROUNDING`, `DETAILED_QA`, and `COARSE_SUMMARY`. Spatially sensitive tasks can therefore be handled using stronger topology preservation than tasks where only coarse visual information is required.

B. Stage 2: FastV Diagnostic

The diagnostic stage examines the scalar attention distribution and evaluates spatial fragmentation. The reported workflow uses visual attention entropy and SCS as indicators of whether scalar attention produces an unreliable or fragmented retention mask.

C. Stage 3: Spatial Graph Regularization

The graph engine constructs the Gaussian affinity matrix and graph Laplacian. The resulting spatially smoothed attention signal is used to reduce local score fluctuations and improve the continuity of retained visual regions.

D. Stage 4: Layer-K Policy Optimization

The policy optimizer evaluates whether pruning can be moved to an earlier Transformer layer. The reported configuration shifts the pruning point from Layer 12 to Layer 8 when the spatially regularized signal is considered sufficiently stable.

E. Stage 5: Multi-Metric Verification

The final stage verifies multiple criteria rather than relying solely on computational savings. The reported metrics include SCS, GQA accuracy, grounding IoU, FLOPs savings, and estimated infrastructure cost.

VI. Experimental Setup

The supplied manuscript reports experiments involving LLaVA-1.5 with a 7B language-model backbone and Qwen-VL-Chat with a 7B backbone. The visual-token configurations are reported as 576 tokens for LLaVA-1.5 and 1024 tokens for Qwen-VL-Chat. The evaluation datasets are reported as Visual Genome for spatial grounding, GQA for spatial reasoning, and LLaVA-Bench for multimodal evaluation.

The comparison includes a dense unpruned baseline, random uniform dropping, ToMe, standard FastV, and TA-FastV. The principal reported comparison is reproduced in Table I.

A. Reported Results

For LLaVA-1.5, the reported SCS increases from 0.241 with standard FastV to 0.948 with TA-FastV. The corresponding reported GQA accuracy increases from 54.8% to 62.6%, while grounding IoU increases from 46.2% to 78.9%. Reported compute cost decreases from 119.4 GFLOPs to 93.3 GFLOPs, corresponding to 48.75% total FLOPs savings under the manuscript's stated calculation.

For Qwen-VL-Chat, the reported SCS changes from 0.265 with FastV to 0.952 with TA-FastV. GQA accuracy is reported as 56.1% for FastV and 64.2% for TA-FastV, while grounding IoU is reported as 48.1% and 81.4%, respectively. The reported FLOPs savings increase from 34.38% to 50.31%.

VII. Discussion

The reported results suggest that explicitly encoding two-dimensional spatial relationships can reduce the fragmentation associated with independent scalar attention ranking. The main difference between FastV and TA-FastV is the way the retained tokens are selected. FastV relies on the attention ranking, whereas TA-FastV also considers the scores of nearby patches.

The graph gives nearby patches a way to influence one another. This can matter in spatial grounding because an object is usually represented by several neighboring patches rather than by one isolated patch. As a result, a patch with a moderate attention value can receive additional support when the surrounding region is also important.

The control workflow is intended to make the pruning decision less rigid. A question that asks for the location of an object may require more spatial information than a question asking only for a broad description of an image. The workflow therefore allows the pruning settings to change with the task instead of using one fixed configuration for every input.

The proposed method is training-free according to the supplied formulation: the graph regularization operates on the attention scores generated during inference and does not require modification of pretrained model weights.

VIII. Limitations and Reproducibility Considerations

The reported results should be interpreted within the exact experimental configuration used to obtain them. For journal submission, the following details should be explicitly documented in the final experimental release: hardware and software versions, exact model checkpoints, image preprocessing and resolution, prompt templates, number of evaluation samples, random seeds, batch sizes, implementation details of attention extraction, the precise definition and connectivity used for SCS, and the method used to measure GFLOPs.

In addition, the manuscript currently reports aggregate benchmark values but does not provide per-dataset confidence intervals, statistical significance tests, ablation studies, or complete implementation details for the agentic policy optimizer. These elements should be added before claiming broad generalization across VLM architectures.

TABLE I
REPORTED QUANTITATIVE COMPARISON OF VISUAL TOKEN REDUCTION METHODS.

Backbone	Method	Cut Layer	SCS	GQA Acc.	Grounding IoU	Compute Cost	FLOPs Saved
LLaVA-1.5 (7B)	Dense Unpruned	None	1.000	63.8%	84.5%	182.0 GFLOPs	0.00%
LLaVA-1.5 (7B)	Random Uniform	12	0.082	41.2%	28.4%	119.4 GFLOPs	34.38%
LLaVA-1.5 (7B)	ToMe	1–32	0.512	57.4%	59.1%	135.2 GFLOPs	25.70%
LLaVA-1.5 (7B)	Standard FastV	12	0.241	54.8%	46.2%	119.4 GFLOPs	34.38%
LLaVA-1.5 (7B)	**TA-FastV**	**8**	**0.948**	**62.6%**	**78.9%**	**93.3 GFLOPs**	**48.75%**
Qwen-VL-Chat (7B)	Standard FastV	12	0.265	56.1%	48.1%	203.4 GFLOPs	34.38%
Qwen-VL-Chat (7B)	**TA-FastV**	**8**	**0.952**	**64.2%**	**81.4%**	**153.9 GFLOPs**	**50.31%**

The cost-reduction estimate should likewise be accompanied by the assumed cloud hardware, pricing model, request volume, token-generation length, and utilization assumptions. Without these details, infrastructure cost should be treated as an estimate rather than a directly reproducible benchmark metric.

IX. CONCLUSION

This work described TA-FastV, a visual-token pruning method that adds a spatial graph to the attention-based selection used by FastV. The graph is used to smooth the attention signal so that neighboring image patches are considered together during pruning. A five-stage control workflow was also included to choose and check the pruning configuration.

For the experimental setup reported in the manuscript, the LLaVA-1.5 results show an SCS increase from 0.241 to 0.948, with pruning moved to Layer 8. The same setup reports 48.75% FLOPs savings and an increase in grounding IoU from 46.2% to 78.9%. These results indicate that spatial information can be useful when deciding which visual tokens to remove. Further experiments, ablations, and independent reproduction are still needed to establish how consistently the method behaves across models and datasets.

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